Mobilization and Collection of HSC

16

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16.1 Introduction

The intravenous infusion of HSC to restore BM damage is the basic principle of high-dose chemotherapy, since otherwise the patient would expect long-lasting aplasia with life-threatening infections. Therefore, a sufficient collection of HSC before application of high-dose therapy is mandatory. Since HSC expresses CD34 on their surface, the number of CD34+ cells in the transplant material is considered the major indicator of the HSC content.

In principle, there are two ways how to collect stem cells: under anesthesia by repeated aspiration of BM from the pelvic crest or by leukapheresis after mobilization of HSC into the PB. The latter one is favored and considered as standard because it is less stressful for the patient or donor and leads to faster engraftment and hematologic reconstitution which may improve patient outcomes (Gertz 2010).

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Usually, HSC circulates in a very small number in the PB (<0.05% of the leukocytes). For HSC mobilization from the BM to the PB agents which stimulate CD34+ cells are needed. For this purpose the cytokine granulocyte colony-stimulating factor (G-CSF) represents the "golden-standard" since several decades. G-CSF induces myeloid hyperplasia and the release of CD34+ cells into the circulation through proteolytic cleavage of adhesion molecules (Lapidot and Petit 2002). Currently, the two G-CSF cytokines filgrastim and lenograstim have market approval for mobilization of HSC in Europe. In case of sufficient mobilization, HSCs are collected by leukapheresis, which is preferentially performed by peripheral venous access or the use of central venous catheters where necessary. Finally, in case of autologous transplantation, the autograft will be cryopreserved using dimethyl sulfoxide (DMSO) until transfusion.

The aim of infusion of HSC from a donor for allogeneic SCT is to restore BM damage and to treat the patient's disease. It represents a permanent cellular immunotherapy by adding a graft versus tumor effect in malignant diseases.

16.2 Strategies of Autologous Stem Cell Mobilization

There are two different strategies to mobilize autologous HSC from the BM to the PB: the so-called steady-state mobilization and the mobili-

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zation by chemotherapy. Both approaches have specific advantages and disadvantages, but the relapse rate is comparable, as documented in several clinical trials (Tuchman et al. 2015).

16.2.1 Mobilization Without Chemotherapy ("Steady State")

Using this approach, HSC will be mobilized by the use of cytokines only. The only approved cytokine for mobilization is G-CSF. The approved doses for steady-state mobilization are filgrastim 10 µg (1.0 Mio. I.E.)/kg/day SC for 5–7 consecutive days or lenograstim 10 µg (1.28 Mio. I.E.)/kg/day SC for 4–6 consecutive days. The use of biosimilar G-CSF has equivalent efficacy (Schmitt et al. 2016; Lisenko et al. 2017).

Leukapheresis usually is performed on day 5 independent whether filgrastim or lenograstim was used for mobilization. If the number of cells collected is inadequate, mobilization with G-CSF may be continued for 1–2 days. However, if the collection goal is not reached after the third leukapheresis, a successful mobilization is unlikely.

The major advantages of steady-state mobilization are the relatively low toxicity including reduced in-hospital patient days, the predictable time of leukapheresis, the outpatient administration, and the reduced costs compared to chemo-mobilization. The major disadvantages are variable mobilization failure rates, and the lower CD34+ cell yields compared to chemo-mobilization. Mobilization with G-CSF only may be used in patients without further need of chemotherapy, e.g., in patients with a stable remission of the underlying disease or patients with multiple myeloma.

16.2.2 Mobilization with Chemotherapy

The use of chemotherapy in combination with G-CSF is the preferred way of mobilization for patients who will need further decrease of tumor burden. CY in a dose of 2 g/m^2 is widely used for HSC mobilization. However, a higher dosage of 4 g/m^2 CY leads to an increased toxicity (Baertsch et al. 2017). In multiple myeloma patients, chemo-mobilization with low-dose cyclophosphamide (2 g/m^2) is a safe mobilization regimen with stem cell collection rates comparable to that of high-dose cyclophosphamide (4 g/m^2) (Zannetti et al. 2021).

It is also an option to mobilize HSC not by a separate mobilization chemotherapy but as part of the disease-specific chemotherapy, e.g., to mobilize HSC following salvage treatment in lymphoma patients. The choice of a specific chemo-mobilization approach is based on patient's disease characteristics and local clinical practice guidelines.

The officially approved doses of G-CSF for HSC mobilization after myelosuppressive therapy are filgrastim 5/µg (0.5 Mio. I.E.)/kg/day SC and lenograstim 150/µg (19.2 Mio. I.E.)/m^2/day SC. In clinical practice, the dosage is often rounded up to 5–10/µg/kg/day filgrastim or 150–300 µg/m^2/day lenograstim, respectively. Mobilization with G-CSF should start after completion of chemotherapy at the earliest and at the leukocyte nadir at the latest and should continue until the last leukapheresis session. Most protocols recommend the initiation of G-CSF within 1–5 days after the end of chemotherapy.

The major advantage of adding chemotherapy to cytokines, besides the effect on the tumor, is the expected improvement of the collection yield with fewer apheresis sessions (Sung et al. 2013). The major disadvantages of chemo-mobilization are the therapy-related toxicity, the requirement of in-hospital treatment in most cases, the bone marrow damage by the chemotherapy which may impair future mobilizations, and higher mobilization costs. Therefore chemo-mobilization may not be the approach of first choice in situations where chemotherapy is not required for the underlying disease.

16.3 CD34+ Cell Count and Timing of Leukapheresis

Up to date, CD34+ cell count in mobilized peripheral blood products is the most important parameter of graft quality, as it is the only

Table 16.1 Recommended start of G-CSF and start of CD34+ monitoring for most used mobilization chemotherapy regimens

Chemotherapy	Start G-CSF	Start CD34+ monitoring
CY 2 g/m^2	Day 5	Day 10
CAD	Day 9	Day 13
(R)CHOP/ CHOEP	Day 6	Day 11
(R)DHAP	Day 9	Day 14
(R)ICE	Day 6	Day 12
(R)AraC/TT	Day 5	Day 10

Day 1: First day of chemotherapy application (without rituximab). Adapted from (Kriegsmann et al. 2018)

recognized predictor of stable hematopoietic engraftment after auto-HCT (Saraceni et al. 2015). Following chemotherapy, an exact prognosis of the CD34+ cell peak in the PB and the optimal start of leukapheresis are difficult and require daily monitoring of CD34+ cells in the PB. Table 16.1 summarizes a recommendation of timing of G-CSF following most of the currently used chemotherapy regimens and start of monitoring of CD34+ cells in the PB.

If not otherwise specified by the protocol, CD34 monitoring should be initiated at the latest if leukocytes increase up to 1000 µL during recovering from aplasia or at day 4–5 of G-CSF in steady-state mobilization. A start of leukapheresis is indicated when a CD34+ cell count of >20/µL is reached (Mohty et al. 2014).

16.4 Target HSC Collection Count

The target number of HSC to be collected is dependent among others on the kind of transplant (e.g., autologous or allogeneic), the stem cell source (BM or PB), and the underlying disease (Table 16.2).

16.4.1 Cell Target for Autologous Transplantation

The generally accepted minimum CD34+ cell yield to proceed to transplantation is 2 × 10^6 cells/kg bw (Mohty et al. 2014). However, higher yields of 2.5–5 × 10^6 CD34+ cells/kg bw have been associated with faster hematopoietic recovery, reduced hospitalization, blood transfusions, and antibiotic therapy (Stiff et al. 2011; Giralt et al. 2014). It may be advantageous to collect cells in this range, if the mobilization status of the patient allows it without additional leukapheresis session.

Table 16.2 Overview on CD34+ dosage

	Autologous transplantation	Allogeneic transplantation
Minimum dose of mobilized CD34+ cells for one transplant	2 × 10^6 CD34+ cells/kg bw	4 × 10^6 CD34+ cells/kg bw
Optimal dose of mobilized CD34+ cells for one transplant	2.5–5 × 10^6 CD34+ cells/kg bw	5–8 × 10^6 CD34+ cells/kg bw[a]

[a] Or higher when further processing is intended

Most patients with NHL or HL will need one autograft. Depending on their risk stratification and the therapeutic protocol, patients may have the need of two or even more transplantations (mainly patients with MM). In these cases, it is essential to collect the required number of HSC before the first high-dose therapy since mobilization after high-dose therapy has an increased risk of failure. For tandem transplantation, the required cell dose for one transplantation is also at least 2 × 10^6 CD34+ cells/kg bw. With new therapies for certain hematological malignancies, such as chimeric antigen receptor (CAR) T cells, the need for tandem transplantation in MM patients may decrease as CAR T cells become widely available.

16.4.2 Cell Target for Allogeneic Transplantation

In allogeneic transplantation, the target quantity of HSC to be collected depends among others on the graft source, the HLA-compatibility between donor and recipient (HLA identical, HLA-haploidentical), the underlying disease (malignant, non-malignant), and the planned graft processing (T-cell depletion, cryopreservation). The generally accepted minimum CD34+

cell dose is ≥4 × 10^6 cells/kg bw of the recipient, which is associated with optimal engraftment kinetics (Miflin et al. 1997). If further processing is planned, doses of up to 8.0 × 10^6 CD34+ cells/kg bw or even more could be necessary.

However, higher doses of CD34+ peripheral blood stem cells are also associated with increased mortality from cGVHD after allogeneic HLA-identical sibling transplantation (Mohty et al. 2003). The probability of extensive cGVHD at 4 years was 34% in patients receiving a CD34+ cell dose <8.3 × 10^6/kg bw, as compared to 62% in patients receiving >8.3 × 10^6/kg bw CD34+ ($P = 0.01$).

16.5 Leukapheresis

Collection of peripheral HSC by leukapheresis is a well-established process. In the allogeneic setting, the duration of one leukapheresis session should not exceed 5 h and two consecutive sessions.

In the autologous setting, some centers perform large volume apheresis with more than 4 times total blood volume to be processed. CD34+ cell collection has been shown to be more effective with larger apheresis volume (4.0–5.3 times the patient's total blood volume), and no difference in CD34+ cell viability was observed compared with normal-volume apheresis (2.7–3.5 times the patient's total blood volume) (Abrahamsen et al. 2005). The duration of one leukapheresis session should still be within the range of 5 h (or within the maximum recommended run time by the manufacturer, i.e., 480 min. for the Spectra Optia), and the total number of leukapheresis sessions should not exceed three consecutive procedures since more sessions are futile in most cases.

To estimate the necessary duration of the leukapheresis session, it may be helpful to use an algorithm based on the CD34+ cell number in PB of the donor and the collection efficiency (CE2) (Wuchter et al. 2017; Rajsp et al. 2022; Kayser et al. 2023).

16.6 Management of Poor Mobilizers

16.6.1 Mobilization Failure Among Patients (Autologous SCT)

Despite widespread and established practice, current mobilization strategies vary between centers and differ in terms of feasibility and outcome. The majority of patients are able to mobilize sufficient CD34+ cells for at least a single auto-HCT. Although the incidence of mobilization failure among patients is not fully documented, approximately 15% of the patients fall in this category (Wuchter et al. 2010). A second attempt to remobilize with G-CSF is usually less effective and has a lower success rate.

Poor mobilizers are defined as patients with less than 2 × 10^6 CD34+ cells/kg collected or patients mobilizing less than 20 CD34+ cells/μL into the PB. In general, there are two groups of poor mobilizers: predicted poor mobilizers and proven poor mobilizers (Olivieri et al. 2012).

Plerixafor (AMD3100) is a bicyclam molecule, which reversibly blocks chemokine receptor-4 (CXCR-4), thereby inhibiting binding with its ligand stroma-cell-derived factor-1 (SDF-1). This mechanism results in the release of hematopoietic progenitor cells in the blood circulation (Uy et al. 2008) and makes plerixafor an important tool to overcome poor mobilization. The addition of plerixafor (recommended dose 0.24 mg/kg bw/day SC or 20 mg abs. in adult patients up to 83 kg bw) to the mobilization scheme should be considered in case of inadequate mobilization (Worel et al. 2017).

If a patient has below 20 CD34+ cells/μL, plerixafor application should be considered. In the "grey area" between 10 and 20 CD34+ cells/μL, the decision to use plerixafor is based on disease characteristics and treatment history. Furthermore, if it is not possible to collect at least one-third of the collection goal with the first apheresis session, plerixafor should be applied as a rescue strategy because of high risk of mobilization failure (Mohty et al. 2014; Cheng et al. 2015). Below 10 CD34+ cells/μL, the use of plerixafor is clearly indicated to avoid mobiliza-

tion failure. Plerixafor is recommended when the CD34+ is <10/μL on days 4–5 of mobilization with G-CSF alone. If the CD34+ is still <10/μL in patients with chemo-mobilization after 12–14 days, plerixafor is recommended if the leukocyte count is increasing (Bilgin 2021).

With the use of plerixafor, patients spend less time on apheresis with less blood volume to be processed and higher CD34+ cell yields with the first apheresis, leading to a decreased number of apheresis sessions needed. This has a direct effect on reducing mobilization costs (Hundemer et al. 2014; Mohty et al. 2018). In case of a failed first mobilization attempt, the use of plerixafor for remobilization is possible and may well be effective (Hubel et al. 2011; Yuan et al. 2013).

Predicted poor mobilizers are defined by baseline patient or disease characteristics which are associated with poor mobilization. These factors are listed in Table 16.3. In patients with one or more of these risk factors, the preemptive use of plerixafor should be considered (Worel et al. 2017). It is generally accepted that the most robust predictive factor for poor mobilization is the CD34+ cell count in PB before apheresis. Patients with low premobilization platelet count (<140 × 10^9/L), age > 65 years and previous radiotherapy were significant predictors of mobilization failure with plerixafor. Also patients who received fludarabine- and lenalidomide-based induction treatment may require plerixafor more often (Bilgin et al. 2015). Nowadays daratumumab, a monoclonal CD38 antibody, is administered to transplant-eligible newly diagnosed myeloma patients and these patients required plerixafor significantly more often (Chhabra et al. 2023).

16.6.2 Mobilization Failure in Allogeneic HSC Donors

In healthy donors, mobilization failure with G-CSF is uncommon, with an estimated incidence rate between 5% and 10% (Ings et al. 2006).

Table 16.3 Factors described as predictive of poor mobilization or mobilization failure

Risk factors for poor mobilization
Age > 60 years
Refractory or advanced stage of underlying disease
High number of prior treatment lines
Therapy with fludarabine, melphalan, lenalidomide and daratumumab
Low CD34+ cell count before apheresis
Low platelet count before mobilization

Modified from (Mohty et al. 2014)

The use of plerixafor is not approved for allogeneic SCT. However, since 2011 several reports mentioned that donors had a successful mobilization using plerixafor after poor mobilization and failure to mobilize adequate CD34+ cell numbers with G-CSF. In several studies, plerixafor was added to the mobilization scheme, with approximately three-fold increase in CD34+ cells (Cid et al. 2021; Hölig et al. 2021).

16.7 Future Directions

At this time, the number of CD34+ cells in the graft is the most important indicator for graft quality. A sufficient number of CD34+ cells are essential to overcome the toxicity of high-dose chemotherapy and to facilitate hematopoietic recovery. However, there is an increasing understanding that other graft subsets, e.g., CD34+ subpopulations or immune cell subsets (B cells, T cells, NK cells, dendritic cells), influence immune recovery. There are also reports that the mobilization regimen has a major impact on graft immune composition and patient's outcome (Saraceni et al. 2015). Therefore, stem cell mobilization could not only be an important part of high-dose therapies but could also be part of an effective immunotherapy. The delineation of this approach has just been started.

More recently, in 2023, the first biosimilar of plerixafor has been approved by the EMA. The impact of this decision is yet unclear regarding marketing price and availability.

Key Points

- Mobilization with chemotherapy plus G-CSF is the preferred method for patients who will need decrease of tumor burden or who have to collect a high number of HSC.
- Steady-state mobilization should be considered for patients who are not in need for chemotherapy due to the underlying disease (e.g., MM) or disease status (i.e., stable remission).
- Up to date, CD34+ cell count in the PB is the most important parameter of graft quality.
- The required dose for one autologous transplant is at least 2×10^6 CD34+ cells/kg bw.
- The required dose for one allogeneic transplant is at least 4×10^6 CD34+ cells/kg bw.
- The indication for the use of plerixafor depends on the CD34+ cell count in the PB, the collection goal, the collection yield with the first apheresis, and the presence of risk factors.

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